Byte Latent Mamba With State Space and Knowledge Distillation for Hyperspectral Image Classification

Muhammad Ahmad, Manuel Mazzara, Salvatore Distefano, and Adil Mehmood Khan

***Abstract*—Hyperspectral image classification (HSIC) is a challenging task due to the high dimensionality of hyperspectral data, the complex interplay of spatial and spectral features, and the scarcity of annotated samples. Existing approaches, mainly based on tokenization-based feature extraction, introduce artificial segmentation, increasing computational cost, and may lead to information loss. To address these issues, a novel framework, byte latent Mamba with knowledge distillation (BLM-KD), which overcomes explicit tokenization by directly learning byte-level spectral–spatial representations from raw hyperspectral data, is proposed. The byte latent Mamba architecture learns compact and expressive byte-level features through an end-to-end convolutional encoder, preserving spectral continuity and spatial structure. A structured state-space model (SSM) is integrated to model long-range spatial–spectral dependencies efficiently via learned dynamic state transitions. Additionally, an adaptive knowledge distillation (KD) strategy is adopted, where a high-capacity teacher model selectively transfers salient features to a lightweight student model, driven by a temperature-controlled weighting schedule. This ensures robust generalization with reduced model complexity. A patch-based preprocessing scheme also excludes irrelevant zero-labeled samples, refining the training process. Extensive experiments conducted on multiple real-world hyperspectral benchmarks demonstrate that BLM-KD outperforms existing state-of-the-art methods in both classification accuracy and computational efficiency.**

***Index Terms*—Byte latent Mamba, hyperspectral image classification (HSIC), knowledge distillation (KD), state-space model (SSM).**

I. Introduction

HYPERSPECTRAL sensors capture extensive spectral information across a wide range of electromagnetic wavelengths, offering superior resolution compared to traditional imaging techniques [1]. Hyperspectral images (HSIs) record continuous spectra across several narrow spectral bands, enabling precise material identification and detailed analysis of Earth surface characteristics [2], [3]. This capability is critical for different applications, including agriculture [4], [5], forestry [6], urban planning [7], [8], environmental monitoring [9], mineral exploration [10], forensic bloodstain identification [11], [12], meat processing [13], [14], contamination detection in red chili [15], [16], and quality assessment of bakery products [17].

HSIs enable the differentiation of materials with similar visual characteristics, such as color and texture, by leveraging their distinct spectral properties [18], allowing for more accurate land-cover classification by overcoming the limitations of conventional imaging methods. Accurate classification is vital for identifying land-cover types, understanding land-use patterns [19], and monitoring environmental changes such as deforestation and urban expansion [20]. Such applications support conservation and resource management strategies.

With the rapid advancements in deep learning (DL), HSI classification (HSIC) has undergone significant improvements [21]. Convolutional neural networks (CNNs) have emerged as a dominant approach for HSIC, capturing both spatial and spectral dependencies in hyperspectral data [22], [23]. However, CNN-based approaches struggle to model long-range dependencies and global contextual relationships, leading to suboptimal performance in complex HSIC tasks [24], [25].

To address these challenges, researchers have explored Transformer-based models leveraging self-attention mechanisms to capture long-range spectral dependencies [26], [27], [28]. While these models enhance feature representation and classification accuracy, they introduce computational overhead due to their quadratic complexity [29], [30], [31]. Recently, state-space models (SSMs), such as the Mamba model, have been proposed as a computationally efficient alternative [32]. The Mamba model leverages hidden state transitions to model spectral dependencies, making it a promising approach for HSIC [33]. Several variants of the Mamba model, such as MHSSMamba [34], WaveMamba [35], and MorpMamba [36], have introduced enhancements like multihead self-attention and wavelet transformations to improve classification accuracy. Despite these advancements, challenges related to scalability and efficiency remain, motivating further research into optimizing Mamba-based models for HSIC.

Meanwhile, knowledge distillation (KD) has been proposed as an effective technique to improve the generalization ability and efficiency of DL models, particularly in settings where

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Muhammad Ahmad is with SDAIA-KFUPM, Joint Research Center for Artificial Intelligence (JRCAI), King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia (e-mail: mahmad00@gmail.com).

Manuel Mazzara is with the Institute of Software Development and Engineering, Innopolis University, 420500 Innopolis, Russia (e-mail: m.mazzara@innopolis.ru).

Salvatore Distefano is with the Dipartimento di Matematica e Informatica-MIFT, University of Messina, 98121 Messina, Italy (e-mail: sdistefano@unime.it).

Adil Mehmood Khan is with the School of Computer Science, University of Hull, HU6 7RX Hull, U.K. (e-mail: a.m.khan@hull.ac.uk).

Data is available on-line at https://github.com/mahmad000/Byte-Latent-Mamba/tree/main

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TABLE I

INNOVATIONS OF THE PROPOSED BLM-KD FRAMEWORK IN COMPARISON WITH EXISTING METHODS, HIGHLIGHTING ADDRESSED LIMITATIONS AND IMPACT ON HSIC

Component	Limitations of existing works	BLM-KD	Impact on HSIC
Byte Latent Representation	Tokenization-based pipelines introduce spatial fragmentation and preprocessing complexity [25], [37].	Learns spectral-spatial byte-level encodings directly from raw input, eliminating the need for Tokenization.	Reduces preprocessing cost, preserves continuity, and improves classification granularity.
SSM	CNN captures only local dependencies; the Transformers model captures global features but with high memory usage [29], [31], [38].	Integrates lightweight structured recurrence to jointly model spectral-spatial contexts with linear time complexity.	Enables efficient long-range modeling and reduces computational load.
Adaptive KD	Existing KD approaches use static logits and lack dynamic adjustment; often inefficient for spectral-spatial domains [39]–[41].	Uses a temperature-controlled, teacher-guided distillation strategy tailored for HSIC-specific feature transfer.	Improves the generalization of the student model while maintaining a lightweight structure.
Scalable Training	High-capacity models risk overfitting and require extensive memory, limiting generalization across datasets [21], [42].	Incorporates regularization (L2, dropout) and balanced parameterization to stabilize training across scales.	Ensures generalization and scalability for large and small datasets.

model size and computational resources are constrained [39]. In the context of HSIC, KD is used to transfer knowledge from a teacher model (typically large and computationally intensive) to a student model (smaller and more efficient) [40]. This technique helps improve the performance of smaller models while maintaining a reduced computational footprint [41]. The idea is to distill the knowledge from a large, complex teacher model (such as a hybrid CNN–Transformer model) into a compact student model, which maintains high classification accuracy while being computationally more efficient [43]. This approach has shown promising results in reducing the resource requirements of HSIC [35]. By distilling knowledge from larger models, KD helps in improving the performance of smaller models without requiring a large labeled dataset, which is often a limitation in HSIC [42], [44].

However, a shortcoming is the reliance on a teacher model, which can be computationally expensive and may require substantial resources to train, especially for high-dimensional hyperspectral data. Additionally, KD models may struggle to capture fine-grained spectral–spatial relationships in HSIs, potentially leading to a loss of important contextual information during the distillation process. Moreover, the effectiveness of KD is often dependent on the alignment between the teacher and student models, which can be challenging to achieve in complex tasks like HSIC. While DL-based approaches such as CNN, Transformers, and Mamba models have significantly improved HSIC performance, they exhibit inherent challenges in modeling long-range dependencies, computational efficiency, and scalability (see Table I and Fig. 1 for a comparative summary).

To address such challenges, this paper proposes the byte latent Mamba with KD (BLM-KD) framework. BLM-KD introduces several key innovations that overcome the limitations of traditional tokenization-based methods and computationally intensive architectures in HSIC. The main contributions of this work are as follows.

1) *BLM Learning:* A tokenization-free spectral–spatial encoding paradigm that eliminates the need for fixed-size tokenization is proposed, allowing the model to directly learns compressed latent representations from raw hyperspectral data via convolutional byte encodings.

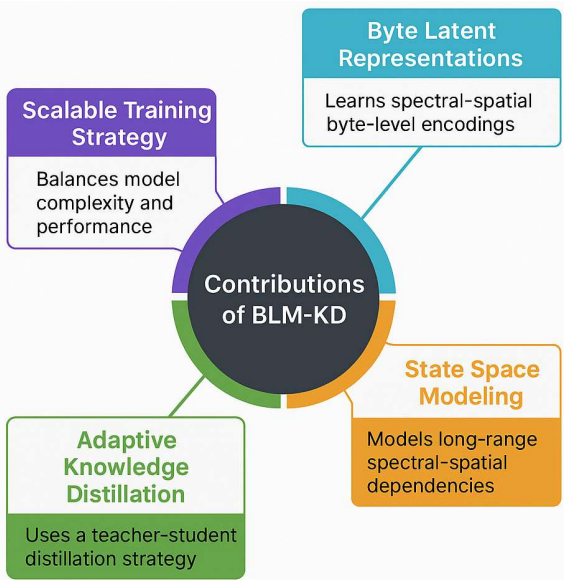


Fig. 1. BLM-KD contribution.

This design preserves spectral continuity and spatial coherence, avoids tokenization artifacts, and reduces computational redundancy.

2) *Integration of the SSM:* The framework incorporates a structured SSM that captures long-range spatial–spectral dependencies using dynamic state transitions. Unlike traditional SSM applications focused solely on spectral sequences, our formulation jointly learns spatial and spectral dynamics, enabling more expressive and adaptive feature representations with linear computational complexity.
3) *Adaptive KD:* An efficient teacher–student distillation scheme is proposed wherein the teacher selectively transfers salient spectral–spatial features. The student model is guided through a dynamic distillation process, reinforced by a temperature-controlled schedule that improves generalization with minimal parameter overhead, tailored specifically for hyperspectral learning tasks.
4) *Scalable and Regularized Training Methodology:* A balanced training strategy is designed to ensure stable

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convergence across varied dataset sizes and resolutions. The use of dropout, L_2 regularization, and warm-up-based KD scheduling mitigates overfitting and maintains performance scalability, making the framework suitable for both low-data regimes and real-time deployment.

II. Literature Review

DL approaches have revolutionized HSIC by enabling automated feature extraction and representation learning. CNN-based methods have been widely used due to their ability to capture spatial and spectral dependencies. CNNs remain strong baselines for hyperspectral feature extraction. Early 2-D CNNs treat the spectrum as stacked channels and learn spatial filters that implicitly fuse bands, whereas 3-D CNNs convolve jointly over (H, W, C) to capture local spectral–spatial patterns more explicitly; separable and lightweight variants reduce parameters by factorizing spatial and spectral kernels. These models provide locality and translation equivariance, but their effective receptive field and spectral kernel width must grow with depth to capture long-range dependencies, which increases computation and memory [45], [46].

For example, the S3F2Net [47] integrates CNNs for local feature extraction with graph convolutional networks (GCNs) to model spatial relationships from LiDAR data. Additionally, a dynamic node updating strategy enhances adaptability to complex topologies. While CNNs have demonstrated remarkable performance, they struggle with capturing long-range dependencies, leading to the adoption of Transformer-based models. The self-attention mechanism in Transformers enables them to model nonlocal dependencies effectively, making them well-suited for HSIC. Their ability to model nonlocal dependencies and provide interpretability through attention mechanisms makes them ideal for hyperspectral data analysis [48], [49]. Moreover, their capacity to capture spectral–spatial dependencies, handle long-range dependencies, and generate contextualized representations makes them a promising approach for extracting discriminative features from hyperspectral data and improving classification accuracy.

Several Transformer-based models have been introduced for HSIC, each incorporating unique architectural enhancements. For instance, Qing et al. [50] proposed SAT Net, integrating self-attention and spectral attention to improve spectral–spatial feature representation. He et al. [51] developed the SST framework, which combines CNNs for spatial feature extraction with a dense Transformer to capture spectral correlations. Scheibenreif et al. [52] enhanced Transformer-based HSIC using self-supervised masked image reconstruction, incorporating block-wise patch embeddings and spectral–spectral self-attention for improved classification accuracy. Qin and Zhao [53] introduced a Transformer framework leveraging spatial–spectral-associative contrastive learning to extract high-level semantic features, mitigate class imbalance, and integrate cross-domain features.

Other works focus on hybrid architectures. Ma et al. [54] introduced the NEHT network, which combines CNNs and Transformers to balance spectral dependency modeling and computational efficiency. Dang et al. [55] developed the DBFFT model, which employs two attention modules for adaptive spectral–spatial feature extraction. Tan et al. [56] proposed an end-to-end classification framework combining DL, endmember extraction, and Transformer-in-Transformer (TNT) modules. Hong et al. [57] proposed a Transformer backbone for HSIC, featuring groupwise spectral embedding (GSE) for spectral differences and cross-layer adaptive fusion (CAF) for enhanced information flow. He et al. [58] introduced Spa-Spe-TR, a dual-branch Transformer where Spa-TR and Spe-TR extract spatial and spectral features, respectively, with branch fusion for joint feature learning.

Despite the advances in Transformer-based methods, their reliance on fixed-size patch tokenization can overlook fine-grained details, and their quadratic complexity poses scalability challenges [31]. Furthermore, while spatial–spectral Transformers (SSTs) capture contextual dependencies well [59], [60], they require substantial labeled data and computational resources, limiting their practical application in HSIC [61].

To address these limitations, SSMs have emerged as an efficient alternative [62]. The Mamba model, in particular, offers significant advantages over traditional Transformers by leveraging state-space representations, which inherently reduce computational demands [63]. Unlike Transformers, which exhibit quadratic complexity due to self-attention mechanisms, Mamba processes spectral sequences linearly, making it more suitable for HSIs with lengthy spectral dimensions [64]. Additionally, Mamba models require fewer parameters and consume less memory, allowing efficient handling of large datasets while preserving performance [65].

For instance, MiM [66] utilizes multiscale learning and feature tokenization to improve classification accuracy while addressing computational bottlenecks. Despite their advantages, Mamba-based models still face scalability challenges, particularly for high-dimensional hyperspectral datasets. The MambaLG [67] introduces a spectral–spatial adaptive Mamba for HSIC, balancing spatial and spectral feature extraction efficiently. The local enhanced Mamba (LE-Mamba) network [68] offers an efficient solution for HSIC by combining a local enhanced spatial SSM (LES-S6) and central region spectral SSM (CRS-S6) for precise spatial–spectral integration, while the multiscale convolutional gated unit (MSCGU) ensures robust feature aggregation, achieving strong noncausal feature extraction and enhanced classification performance. The HLMamba [69] model leverages a gradient joint algorithm (GJA) for edge contour extraction and a multimodal Mamba fusion module (MMFM) for efficient feature integration across HSI and LiDAR data, enhancing classification performance by capturing complex interrelationships and long-distance dependencies in multimodal data. The spatial–spectral interaction super-resolution CNN–Mamba fusion network [70] enhances the resolution of low-spatial-resolution HSIs and high-spatial-resolution multispectral images through mutual guidance, ensuring effective computational efficiency in extracting both local and global features for improved image fusion.

HSIC has evolved considerably through the use of CNNs, Transformers, and, more recently, SSMs. CNNs are effective at capturing local spatial features due to their strong inductive

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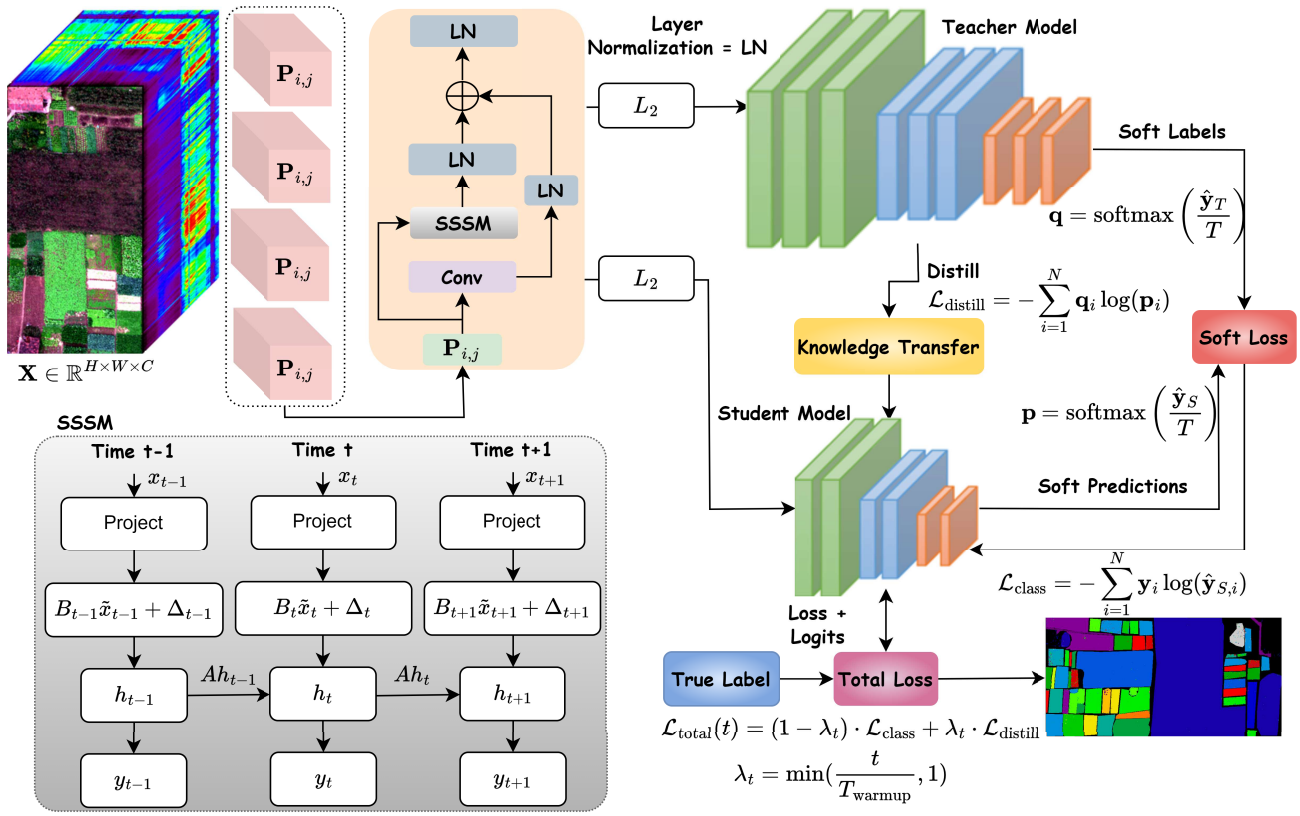


Fig. 2. Overview of the proposed KD framework for HSIC. Hyperspectral patches ($P_{i,j}$) are processed through an SSM, which includes operations like patching, convolutional layers with ReLU activation, and layer normalization (LN). The extracted features are used for both the teacher and student models, where the teacher model supervises the student model using knowledge transfer and L_2-norm-based regularization. Projection, class predictions (hard labels), and final classification losses are utilized for optimization.

biases and fixed receptive fields. However, they struggle with modeling long-range dependencies and tend to overfit when applied to datasets with limited labeled samples. Transformers have introduced the ability to capture global dependencies through self-attention mechanisms, but they suffer from quadratic computational complexity and lack spatial inductive biases. In response, recent work on SSMs such as Mamba introduces a structured linear recurrence that efficiently models long-range interactions with linear-time complexity. Yet, these models are not tailored for spatial–spectral encoding in HSIC and often neglect lightweight feature extraction or student–teacher learning strategies essential for generalization.

To overcome these limitations, we propose the BLM-KD framework, which introduces several key innovations. At its core, BLM-KD incorporates a tokenization-free byte latent representation (BLM) learning mechanism that directly operates on the raw hyperspectral patch cube without tokens. Existing tokenization-based methods lead to information fragmentation and inefficiencies in feature representation. In contrast, our model learns compressed, end-to-end optimized latent encodings that preserve hierarchical spectral dependencies and spatial continuity. This approach not only avoids preprocessing artifacts but also enhances feature granularity, reduces redundant computation, and improves both classification performance and computational efficiency.

While SSMs have been used in prior HSIC methods, their applications are mostly restricted to modeling sequential spectral dependencies alone. These traditional approaches fail to capture adaptive spatial relationships. BLM-KD extends the use of SSMs by embedding a dynamically structured state transition mechanism that jointly learns spectral transitions and spatial correlations. This integration leads to robust and expressive hyperspectral representations capable of supporting complex classification tasks across varied scenes.

Another important advancement in our framework is the deployment of an optimized KD strategy. Although KD has been explored in general DL tasks, its use in HSIC remains limited due to the difficulty of distilling relevant information across redundant spectral bands and spatial inconsistencies. Our KD formulation adopts an adaptive teacher–student training paradigm in which the teacher selectively distills salient spectral–spatial features. This targeted distillation enhances the efficiency of feature transfer and reduces information redundancy, resulting in a compact, yet highly effective student model. Unlike conventional KD schemes that mainly compress general features, our approach fine-tunes the distillation specifically for hyperspectral data, yielding high classification accuracy with reduced model complexity. In short, the proposed BLM-KD framework addresses core limitations of prior HSIC models by removing the dependency on rigid tokenization, embedding structured recurrence for efficient spatial–spectral modeling, and incorporating a robust distillation mechanism. The complete architecture is presented in Fig. 2.

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III. Proposed Methodology

A. BLM Learning

Let the HSI be represented by a 3-D tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where H and W denote the spatial dimensions (height and width) and C denotes the number of spectral bands. The corresponding ground-truth label map is given by $\mathbf{Y} \in \mathbb{R}^{H\times W}$. To learn discriminative features centered at each pixel while considering its spatial context, we first extract a local 3-D patch $\mathbf{P}_{i,j}$ of size $k \times k \times C$ centered around pixel (i, j). This is mathematically expressed as $\mathbf{P}_{i,j} = \mathbf{X}_{i-\lfloor k/2\rfloor:i+\lfloor k/2\rfloor+1,:j-\lfloor k/2\rfloor:j+\lfloor k/2\rfloor+1,::} \in \mathbb{R}^{k\times k\times C}$ and the associated class label of this patch is $y_{i,j} = \mathbf{Y}_{i,j}$.

This patch-based extraction ensures that the local spatial context is retained along with complete spectral signatures. All extracted patches where $y_{i,j} = 0$ (i.e., background or unlabeled regions) are discarded to focus on meaningful classes.

Let us denote the extracted and batched patch tensor as $\mathbf{X}_{\text{patch}} \in \mathbb{R}^{B\times k\times k\times C}$, where B is the batch size. For compactness, we let $W = k$. We apply a two-layer 2-D convolutional encoder on the spectral–spatial patch to extract intermediate features. Each convolution operation is followed by a ReLU activation

$$\mathbf{F}1 = \sigma\left(\mathbf{X}_{\text{patch}} * \mathbf{K}_1 + \mathbf{b}_1\right), \quad \mathbf{F}_1 \in \mathbb{R}^{B\times W\times W\times C'} \tag{1}$$

$$\mathbf{F}_2 = \sigma\left(\mathbf{F}_1 * \mathbf{K}_2 + \mathbf{b}_2\right), \quad \mathbf{F}_2 \in \mathbb{R}^{B\times W\times W\times C} \tag{2}$$

where $\mathbf{K}_1$ and $\mathbf{K}_2$ are the learnable convolutional kernels, $\mathbf{b}_1$ and $\mathbf{b}_2$ are the bias terms, and $\sigma(\cdot)$ denotes the ReLU activation. The first convolution increases the number of channels to $C' = 2C$ to enrich intermediate features, while the second brings the output back to C channels, which matches the SSM input dimensionality. Dropout regularization is then applied to prevent overfitting

$$\mathbf{F}_{\text{drop}} = \text{Dropout}\left(\mathbf{F}_2\right), \quad \mathbf{F}_{\text{drop}} \in \mathbb{R}^{B\times W\times W\times C} \tag{3}$$

where $\mathbf{F}_{\text{drop}}$ randomly deactivates neurons during training and enhances generalization. To model long-range spatial–spectral dependencies using a sequential SSM, the spatial dimensions are flattened

$$\mathbf{P}_{\text{seq}} = \text{Flatten}_{\text{spatial}}\left(\mathbf{F}_{\text{drop}}\right) \in \mathbb{R}^{B\times T\times C}, \quad \text{where } T = W^2 \tag{4}$$

where each spatial location in the $W \times W$ patch is treated as a time step in a sequence of length T, with C-dimensional input vectors.

Conventional HSI methods often operate on either raw spectral vectors or handcrafted patch-level features. Transformer-based models tokenize spatial regions, while SSMs typically sequence-encode entire patches. Our method departs from both by introducing a BLM, a compressed, learnable embedding of the spectral–spatial cube. Each input patch $\mathbf{P}_{i,j} \in \mathbb{R}^{k\times k\times C}$ is processed through a series of learnable convolutions that mimic the function of byte-level encodings, that is, extracting small, compressed units of information that still preserve both spectral continuity and spatial semantics. Unlike hard-coded token sequences, this representation is optimized end-to-end, allowing the encoder to learn the most discriminative latent units.

Formally, this encoding allows the model to operate over sequences of shape $\mathbb{R}^{B\times T\times d}$, where T is the number of spatial positions and d is the learned embedding dimension. This gives the model flexibility and allows downstream modules to operate on compressed yet rich inputs, improving generalization and computational efficiency.

B. Integration of a Structured State-Space Model

Therefore, the SSM employed in our framework is designed to model long-range dependencies efficiently by leveraging learned dynamics through structured linear recurrence. The model operates sequentially over input and updates its hidden state over time, making it suitable for spatial–spectral data where continuity and memory are crucial. Given the input sequence at time step t, denoted as $x_t \in \mathbb{R}^d$, we first apply a linear projection to map it into a latent space

$$\tilde{x}_t = \text{Project}\left(x_t\right) \tag{5}$$

where $\tilde{x}_t \in \mathbb{R}^{d'}$. This projection helps align the input dimensionality with that of the state model and allows learning a richer representation before integration into the recurrent pipeline. To incorporate sequential context, we inject a learnable time embedding Δ_t into the input. This is done through a learnable matrix $B_t \in \mathbb{R}^{n\times d'}$ that modulates the projected input

$$u_t = B_t\tilde{x}_t + \Delta_t \tag{6}$$

where $u_t \in \mathbb{R}^n$ is the input term to the state dynamics. The addition of Δ_t introduces time awareness, allowing the model to differentiate between inputs based on their sequential position, which is particularly useful for spatial–spectral modeling.

The core of the SSM lies in its recurrence relation. At each time step t, the hidden state $h_t \in \mathbb{R}^n$ is updated using the previous hidden state h_{t-1} and the input term u_t, as follows:

$$h_t = Ah_{t-1} + u_t \tag{7}$$

where $A \in \mathbb{R}^{n\times n}$ is a learnable state transition matrix. This formulation enables the model to capture long-term dependencies through linear evolution of the hidden state while maintaining computational efficiency. Finally, the output at time step t, denoted $y_t \in \mathbb{R}^{d''}$, is computed by applying another linear transformation to the hidden state

$$y_t = D_th_t \tag{8}$$

where $D_t \in \mathbb{R}^{d''\times n}$ is a learnable output projection matrix and y_t represents a compact summary of the spatiospectral information for the central pixel. This allows flexible transformation of the state into the desired output space for classification. The SSM thus combines a memory mechanism (via h_t) with learnable dynamics (via A, B_t, and D_t) and time encoding (Δ_t) to effectively model spatial and spectral dependencies. Compared to conventional recurrent neural networks (RNNs) or Transformers, this formulation allows for linear-time processing while maintaining the ability to capture both short- and long-term correlations. Finally, the encoded vector y_t is passed to a fully connected layer followed by a Softmax activation to produce class probabilities

$$\hat{\mathbf{y}} = \text{softmax}\left(\mathbf{W}_c\mathbf{h}_{\text{final}} + \mathbf{b}_c\right), \quad \hat{\mathbf{y}} \in \mathbb{R}^{B\times N} \tag{9}$$

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where $\mathbf{W}_c \in \mathbb{R}^{N\times D}$ and $\mathbf{b}_c \in \mathbb{R}^N$ are the weight and bias parameters of the classifier, respectively, and N is the number of classes. The Softmax operation normalizes the output into a valid probability distribution over classes.

Theorem 1 (Boundedness and stability of the state sequence): Assume there exists a submultiplicative matrix norm $\|\cdot\|$ such that $\|A\| = \rho < 1$, and suppose $\|B_t\| \le B_{\max}$, $\|\tilde{x}_t\| \le X_{\max}$, and $\|\Delta_t\| \le \Delta_{\max}$ for all t. Then, the hidden state is uniformly bounded as

$$\|h_t\| \le \rho^t \|h_0\| + \frac{B_{\max} X_{\max} + \Delta_{\max}}{1-\rho} \quad \text{for all } t \ge 0. \tag{10}$$

Moreover, the input-to-state map $\{x_t\}_{t=1}^T \mapsto \{h_t\}_{t=1}^T$ is Lipschitz with constant at most $B_{\max}/(1-\rho)$ in the induced sequence norm.

Proof: Unroll the recurrence to obtain $h_t = A^t h_0 + \sum_{k=0}^{t-1} A^{\text{t-1-k}}(B_k \tilde{x}_k + \Delta_k)$. The norm bound follows from the triangle inequality and $\|A^{\text{t-1-k}}\| \le \rho^{\text{t-1-k}}$. Summing the geometric series gives the stated bound. The Lipschitz claim follows from the linearity of the map in the inputs and the same geometric series estimate.

Proposition 1 (Affine-to-Linear Lifting With Time Embeddings): The affine, time-varying system

$$h_t = A h_{t-1} + B_t \tilde{x}_t + \Delta_t \tag{11}$$

is equivalent to a linear system in an augmented input: define $\tilde{x}'_t = \begin{bmatrix} \tilde{x}_t \\ 1 \end{bmatrix}$ and $B'_t = [B_t \;\; \Delta_t]$. Then,

$$h_t = A h_{t-1} + B'_t \tilde{x}'_t. \tag{12}$$

Consequently, any stability property that depends solely on $\|A\|$ is preserved under the addition of Δ_t.

Proof: Substituting the definitions of $\tilde{x}'_t$ and B'_t, we have

$$B'_t \tilde{x}'_t = [B_t \;\; \Delta_t] \begin{bmatrix} \tilde{x}_t \\ 1 \end{bmatrix} = B_t \tilde{x}_t + \Delta_t. \tag{13}$$

Therefore,

$$h_t = A h_{t-1} + B_t \tilde{x}_t + \Delta_t = A h_{t-1} + B'_t \tilde{x}'_t \tag{14}$$

as claimed. Since the recursion for h_t in the lifted system has the same transition matrix A as in the original system, any stability property depending only on $\|A\|$ is preserved after the transformation.□

Remark 1 (Computational Complexity in Sequence Length): For fixed state dimension n and embedding dimension d', the per-step update $h_t = Ah_{t-1} + B_t\tilde{x}_t + \Delta_t$ is $O(n^2 + nd')$, so the total cost scales linearly in T as $O(T(n^2 + nd'))$, which is linear in the sequence length $T = W^2$, contrasting with quadratic attention mechanisms in T.

C. Adaptive KD

To improve generalization and robustness, we apply KD by combining the standard cross-entropy classification loss with a soft distillation loss from a teacher model. Let $\hat{\mathbf{y}}_T$ be the teacher logits and $\hat{\mathbf{y}}_S$ be the student logits

$$\mathbf{q} = \text{softmax}\left(\frac{\hat{\mathbf{y}}_T}{T}\right); \quad \mathbf{p} = \text{softmax}\left(\frac{\hat{\mathbf{y}}_S}{T}\right) \tag{15}$$

where T is the temperature parameter that smooths the probability distributions. The distillation loss is the Kullback–Leibler (KL) divergence between softened outputs

$$\mathcal{L}_{\text{distill}} = -\sum_{i=1}^{N} \mathbf{q}_i \log(\mathbf{p}_i). \tag{16}$$

The standard classification loss is computed as

$$\mathcal{L}_{\text{class}} = -\sum_{i=1}^{N} \mathbf{y}_i \log\left(\hat{\mathbf{y}}_{S,i}\right). \tag{17}$$

To improve the effectiveness of the distillation process, we incorporate an adaptive weighting scheme into the loss function. Let λ_t be a dynamic scalar that varies with training epoch t, allowing the model to gradually shift focus from the teacher's guidance to the ground-truth supervision. The updated total loss becomes

$$\mathcal{L}_{\text{total}}(t) = (1-\lambda_t)\cdot\mathcal{L}_{\text{class}} + \lambda_t \cdot \mathcal{L}_{\text{distill}} \tag{18}$$

where

$$\lambda_t = \min\left(\frac{t}{T_{\text{warmup}}}, 1\right) \tag{19}$$

where T_{warmup} is a warmup period over which distillation becomes more prominent. This schedule encourages the student to rely on the ground-truth early in training, but smoothly integrates soft-label guidance as training stabilizes.

Theorem 2 (Mixture-Target Equivalence of the Weighted Objective): Let $\mathbf{y}$ be a one-hot target on the simplex Δ^{N-1} and $\mathbf{q}$ be the teacher distribution at temperature T. Define $\mathbf{r}_\lambda = (1-\lambda)\mathbf{y} + \lambda\mathbf{q}$. Then, for any student distribution $\mathbf{p}$

$$\begin{aligned}(1-\lambda)\,\mathcal{L}_{\text{class}} + \lambda\,\mathcal{L}_{\text{distill}} &= -\sum_{i=1}^{N} \mathbf{r}_{\lambda,i} \log \mathbf{p}_i \\ &= \text{CE}(\mathbf{r}_\lambda, \mathbf{p})\end{aligned} \tag{20}$$

that is, the weighted sum is exactly the cross-entropy with the convex combination target $\mathbf{r}_\lambda$.

Proof: Substitute the definitions and collect terms

$$\begin{aligned}&-(1-\lambda)\sum_i \mathbf{y}_i \log \mathbf{p}_i - \lambda \sum_i \mathbf{q}_i \log \mathbf{p}_i \\ &= -\sum_i \left[(1-\lambda)\,\mathbf{y}_i + \lambda \mathbf{q}_i\right] \log \mathbf{p}_i.\end{aligned} \tag{21}$$

Because both $\mathbf{y}$ and $\mathbf{q}$ lie on the simplex, $\mathbf{r}_\lambda$ also lies on the simplex, yielding the desired equality.

Lemma 1 (Monotonic Teacher Influence Under Warmup): With $\lambda_t = \min(t/T_{\text{warmup}}, 1)$, the sequence $(\lambda_t)_{t\ge 0}$ is nondecreasing and converges to 1. Hence, the training objective transitions monotonically from $\mathcal{L}_{\text{class}}$ toward $\mathcal{L}_{\text{distill}}$, never decreasing the relative teacher weight after any epoch.

Proof: Immediate from the definition of λ_t as the minimum of a linear function and 1.

Proposition 2 (Gradient Scaling With Temperature): Let $\mathbf{z}$ be the student logits and $\mathbf{p}_T = \text{softmax}(\mathbf{z}/T)$. Then, the gradient of $\text{CE}(\mathbf{q}_T, \mathbf{p}_T)$ with respect to $\mathbf{z}$ equals $1/T(\mathbf{p}_T - \mathbf{q}_T)$. Therefore, higher temperatures smooth $\mathbf{q}_T$ and proportionally scale down gradient magnitudes, which stabilizes early training when combined with the warmup schedule.

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Proof: Differentiate softmax with temperature and apply the standard cross-entropy gradient identity.

Theorem 3 (Lipschitz Continuity From Inputs to Logits): Under the assumptions of Theorem 1 and bounded classifier parameters $\|\mathbf{W}_c\| \le W_{\max}$, the mapping from the input sequence $\{x_t\}_{t=1}^{T}$ to the student logits before Softmax is Lipschitz with constant at most $W_{\max}\|D_t\|B_{\max}/(1-\rho)$ for each time step. Consequently, small spectral–spatial perturbations in $\mathbf{P}_{i,j}$ induce proportionally small changes in the predicted logits, supporting robustness and generalization.

Proof: Combine the Lipschitz input-to-state bound from Theorem 1 with linearity of $y_t = D_t h_t$ and the final affine classifier $\mathbf{W}_c\mathbf{h}_{\text{final}} + \mathbf{b}_c$.

D. Scalable and Regularized Training

Proposition 3 (Dropout Scaling and Implicit Regularization): Consider a linear layer $\mathbf{z} = \mathbf{W}\mathbf{a}$ applied to activations $\mathbf{a}$ that are subjected to element-wise dropout with keep probability $p \in (0, 1]$ and inverted scaling during training. Then, $\mathbb{E}[\mathbf{z}_{\text{drop}}] = \mathbf{W}\mathbf{a}$ and, for a locally quadratic loss $\ell(\mathbf{z}) \approx 1/2\|\mathbf{z} - \mathbf{t}\|_2^2$, the expected objective equals the no-dropout objective plus an ℓ_2 penalty proportional to $(1-p)/p\|\mathbf{W}\|_F^2 \cdot \mathbb{E}\|\mathbf{a}\|_2^2$. Consequently, dropout acts as a data-dependent weight decay that improves generalization.

Proof: Linearity of expectation with inverted scaling yields the unbiasedness. Expanding the quadratic loss under Bernoulli noise and taking expectations introduces a variance term that is proportional to $(1-p)/p$ and $\|\mathbf{W}\|_F^2$, giving the claimed penalty.

The proposed BLM-KD pipeline leverages the rich spectral signatures and local spatial structure by first extracting joint spectral–spatial patches. The use of lightweight convolutional layers ensures effective local feature encoding with low computational cost. The transformation of spatial positions into a sequential form allows the SSM to model long-range dependencies across the spatial domain efficiently. The final classification layer benefits from a global representation that incorporates both spectral richness and spatial coherence. The addition of KD further regularizes the model by aligning the student predictions with a more accurate teacher, enhancing generalization.

IV. Experimental Settings and Datasets

The experimental settings for this study are outlined in detail below, highlighting the key parameters, model architecture, and evaluation methodology. "For the experiments, each HSI was divided into 8×8 patches, yielding N_{patch} labeled samples. These were randomly split into 25% training, 25% validation, and 50% test sets." These ratios ensure that the dataset is appropriately divided for model training, validation, and testing. In this study, we utilize a combination of the convolutional layers and SSM integrated with KD. The architectures used in both the teacher and student models are as follows.

SSM: The model uses a state-space architecture with two key operations: state transition and state update. The transition and update steps are implemented using dense layers with L_2 regularization to capture the state evolution across sequential data. The state dimension for the teacher model is set to 128, and for the student model, it is halved to 64.

Convolutional Layers: Two 5×5 and 3×3 convolution layers are applied to extract features from the input patch. The output of these convolution layers is passed through a dropout layer with a rate of 0.1 to mitigate overfitting.

KD: The KD technique is applied between a teacher model and a student model, where the teacher model is larger and more complex, and the student model is a smaller version trained to mimic the teacher's output. This approach is implemented through a custom loss function combining both classification and distillation losses. The distillation temperature is set to $T = 1.0$.

The key hyperparameters used in this study are as follows.

Learning Rate: The Adam optimizer with a learning rate of 0.001 is employed for training.

Decay: A weight decay of $1e^{-06}$ is applied for regularization to avoid overfitting.

Batch Size: The batch size is set to 16, determining the number of samples processed in each training step.

Epochs: The model is trained for 50 iterations to ensure sufficient learning. The teacher and student models are both initialized with different numbers of output channels, with the teacher model having a larger number of channels compared to the student model. Specifically, the teacher model has out channels teacher = batch size, and the student model has half the number of output channels: out channels student = out channels teacher/2.

The training of the models is conducted using the Adam optimizer with the learning rate and decay as described above. The models are trained using the KS approach, where the teacher model guides the student model's learning by providing soft targets through the distillation loss function. The distillation loss is a combination of two components as follows.

Classification Loss: The standard categorical cross-entropy loss is computed between the true labels and the predictions of the student model.

Distillation Loss: The difference between the teacher model's soft labels and the student model's predictions is calculated using the KL divergence. The total loss is a sum of the classification and distillation losses, ensuring that both the classification accuracy and the teacher's knowledge are preserved in the student model.

The models are evaluated using the following metrics.

Overall Accuracy: Measures the percentage of correct predictions over all samples.

Per-Class Accuracy (PCA): Reports the accuracy for each class in the dataset.

Average Accuracy: Computes the mean accuracy across all classes.

Kappa Score: Measures the agreement between predicted and true labels, adjusted for chance agreement. The ground-truth labels are compared with the model's predictions to generate these evaluation metrics.

To evaluate the performance and robustness of the proposed BLM-KD model, we conducted experiments using multiple benchmark datasets with diverse characteristics such as WHU-

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Hi-HanChuan (HC), WHU-Hi-HongHu (HH) [71], Salinas (SA), Pavia University (PU), and University of Houston (UH) [72]. These datasets encompass a variety of spectral and spatial resolutions, land-cover types, and environmental conditions, providing a comprehensive basis for assessing the model's applicability across real-world scenarios. The selected datasets include both UAV-acquired hyperspectral imagery, which features fine-grained spatial resolution and complex agricultural and urban environments, and well-established airborne datasets with broader coverage and a focus on spectral diversity. By leveraging such diverse datasets, we aim to rigorously test the model's ability to adapt to varying data distributions and application contexts.

V. Ablation Study

To thoroughly evaluate the performance of the proposed BLM-KD model, we conducted an extensive ablation study focusing on key design parameters. The investigation systematically assessed **KD type comparison, BLM versus tokenization, training data proportion, patch sizes**, and **distillation temperature**. These experiments were systematically designed to examine the relationships between each parameter and the model's performance in terms of classification accuracy, computational efficiency, and robustness. The insights gained from the ablation study highlight optimal configurations for HSIC, enabling a balanced tradeoff between performance and computational cost.

A. KD Type Comparison

To assess the influence of different KD strategies on HSIC, we compared four popular KD paradigms: logit-based, feature-based, response-based, and relation-based. As illustrated in Fig. 3, logit-based KD consistently yields the highest OA across all datasets. This result emphasizes its effectiveness in transferring spectral–spatial knowledge, which aligns well with the data characteristics of HSIs. Specifically, logit-based KD outperforms the other KD strategies by a significant margin, reinforcing its efficiency in leveraging the final-layer logits to transfer key information from the teacher model.

Although response-based KD shows moderate improvements in some datasets, it does not surpass the performance of logit-based KD. The feature-based KD, while also competitive, demonstrates slightly lower performance than logit-based KD due to its reliance on lower-layer alignment, which does not fully capture the complex interdependencies between spatial and spectral features. On the other hand, relation-based KD, although it accounts for intersample dependencies, performs less effectively in hyperspectral domains with high-dimensional data, yielding the lowest OA across the datasets. These findings validate the choice of logit-based KD in the BLM-KD pipeline, confirming that it provides the most robust and efficient transfer of structured spectral–spatial knowledge, crucial for HSIC.

B. BLM Versus Tokenization

To evaluate the effectiveness of BLM, we conducted a comparative analysis against conventional tokenization. The results, illustrated in Fig. 4, clearly demonstrate the superiority of BLM in both classification accuracy and computational efficiency. Specifically, the BLM consistently achieved higher OA across all datasets. For example, on the HC dataset, BLM attained an OA of 98.22%, outperforming tokenization, which achieved 96.90%. Similar trends were observed for other datasets, such as PU and SA, confirming the robustness of BLM across diverse spectral–spatial conditions. Moreover, BLM also exhibited significantly lower computational cost, with up to 25%–30% reductions observed across all datasets. This is particularly notable on the SA and UH datasets, where BLM maintained superior OA while reducing resource consumption. The decrease in computational cost is attributed to BLM's direct mapping of raw data into compact byte-level representations, eliminating the need for redundant token generation and associated overhead.

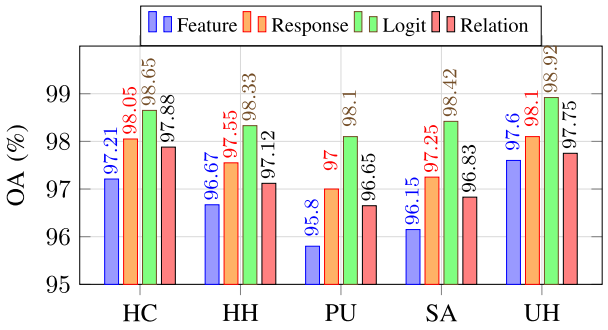


Fig. 3. Comparison of KD strategies on five scenes (HC, HH, PU, SA, and UH). Bars show OA (%). Logit-based KD attains 98.65/98.33/98.10/98.42/98.92 on HC/HH/PU/SA/UH (macroaverage 98.48), outperforming response (97.59), relation (97.25), and feature (96.69) variants under identical training settings.

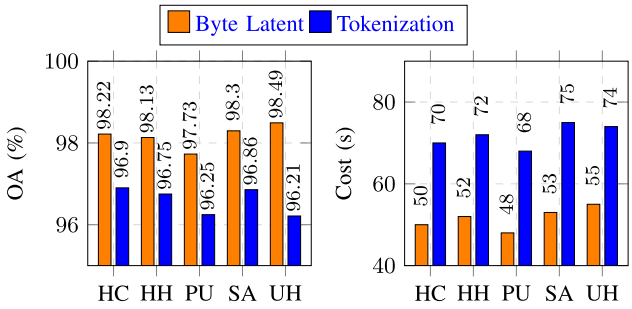


Fig. 4. Byte latent versus tokenization on five scenes (HC, HH, PU, SA, and UH). Macroaverages: OA 98.17 versus 96.59; cost 51.6 versus 71.8 s (~1.39× faster).

C. Distillation Temperature

The distillation temperature is a critical hyperparameter that influences the performance of models designed for HSIC. It plays a key role in KD, where a teacher model transfers knowledge to a student model. Adjusting the temperature impacts the softening of probability distributions during training, thus balancing the tradeoff between preserving high-confidence predictions and capturing interclass similarity.

Fig. 5 illustrates the performance variations of five datasets across different temperature settings. The OA values are plotted for each dataset, highlighting the nuanced effects of temperature adjustments on classification accuracy. For most

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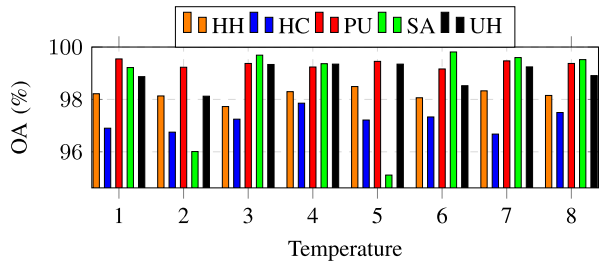


Fig. 5. Performance variations with the distillation temperature for five datasets.

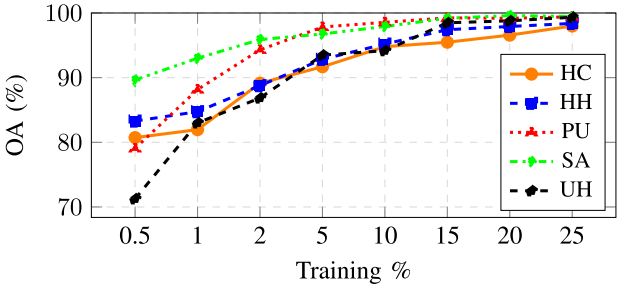


Fig. 6. Influence of training sample percentages on OA, illustrating the model's robustness with small datasets and steady performance gains as the training data increases. The performance of the BLM-KD model with respect to different training data percentages is evaluated using an 8×8 patch size.

datasets, there is a noticeable variation in OA with temperature adjustments. This indicates that the optimal temperature differs based on dataset characteristics, such as spectral and spatial heterogeneity.

PU achieves the highest OA values consistently across all temperatures, peaking at certain intermediate levels. This suggests that PU benefits from robust KD, likely due to its well-defined spectral features. The performance for SA exhibits significant variability, with a sharp drop at specific temperatures (e.g., 4). This highlights the dataset's sensitivity to temperature settings, which could stem from class imbalance or increased spectral mixing. The UH dataset shows relatively stable performance, with minor variations across temperatures. This stability indicates that the model effectively adapts to different temperature settings, likely due to the uniform distribution of classes. The HC and HH datasets demonstrate modest OA values compared to others, suggesting that the distillation temperature needs to be fine-tuned to fully exploit their spectral–spatial characteristics.

Across all datasets, the temperature range between 2 and 6 generally yields higher OA values. This range appears to strike a balance between overly sharp probability distributions (low temperatures) and overly softened distributions (high temperatures). The results emphasize the importance of dataset-specific tuning of the distillation temperature. The significant variations in OA suggest that the spectral–spatial diversity and class distribution of the datasets heavily influence their sensitivity to this parameter.

D. Effect of Training Sample Proportions

Fig. 6 presents a comprehensive analysis of how varying the percentage of training samples influences the OA across multiple datasets. This ablation study highlights the model's robustness and adaptability when exposed to limited training data, a critical attribute for HSIC, where obtaining labeled data can be particularly challenging.

The improvement in OA for the HC dataset is most pronounced in the early stages, particularly between 0.5% and 5% training data. This sharp increase reflects the model's ability to effectively leverage limited samples in datasets characterized by high-class heterogeneity. The UH dataset, in contrast, demonstrates a slower initial rate of improvement, which may be attributed to its relatively uniform spectral properties, allowing the model to generalize more readily even with fewer training samples. For all datasets, the OA begins to plateau beyond 15% training data, highlighting the model's efficiency in learning the underlying spatial–spectral patterns with relatively few samples. This results in diminishing returns from additional training data at higher percentages. The HH dataset showcases smoother transitions, indicating balanced spectral variability and sufficient class representation within the datasets. These results clearly demonstrate the model's capability to achieve high performance even with small datasets. For instance, with only 5% training data, the model achieves an impressive OA of over 90% for all datasets, surpassing many conventional approaches that require significantly larger training sets. At 25% training data, the model achieves near-optimal performance, with OAs exceeding 98% across all datasets, further demonstrating the model's efficacy in both resource-limited and data-abundant scenarios.

This ablation study not only highlights the model's adaptability across diverse datasets but also provides actionable insights into its data efficiency. The findings validate the hypothesis that the proposed model is well-suited for scenarios where labeled data availability is constrained. Such robustness makes it particularly valuable for real-world applications, such as agricultural monitoring, environmental assessment, and disaster response, where acquiring labeled data can be labor-intensive or costly. Moreover, the analysis reveals how dataset-specific characteristics, such as spectral uniformity and class diversity, influence the learning dynamics of the model. Understanding these trends allows for fine-tuning and optimization of the model, further enhancing its utility in domain-specific applications. The study demonstrates that the proposed model not only achieves state-of-the-art performance but does so with remarkable data efficiency. The ablation results provide compelling evidence of the model's capability to generalize effectively, even in challenging scenarios, making it a reliable tool for HSIC tasks.

E. Impact of the Patch Size

The patch size is crucial for understanding the sensitivity of the BLM-KD model. In HSIC, the patch size determines the spatial and spectral context provided to the model. Larger patches capture broader spatial information but may dilute local details, while smaller patches provide focused spatial details but risk missing global context. This tradeoff is especially critical for models leveraging spectral–spatial feature learning, as the patch size directly influences the quality and quantity of extracted features.

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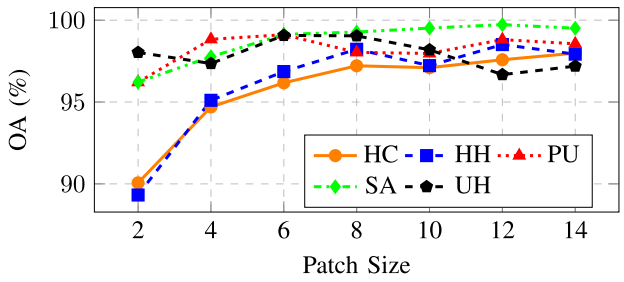


Fig. 7. Impact of the patch size on OA for different datasets.

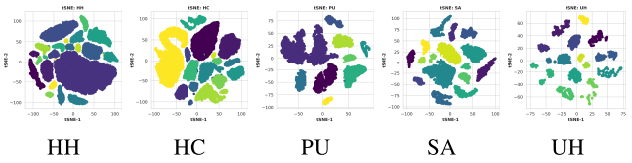


Fig. 8. t-SNE visualization of the learned feature representations for the PU, PC, SA, UH, and HC datasets, showcasing the BLM-KD model's ability to effectively distinguish and cluster features for classification in a lower-dimensional space.

Across datasets, there is a noticeable trend where the classification accuracy improves with increasing patch size up to a certain threshold, after which it saturates or marginally decreases. The trend highlights that an optimal patch size exists, balancing local and global feature representation. HC and HH datasets show a consistent improvement in accuracy up to patch sizes of 12 or 14, reflecting their spatial dependencies. PU and SA datasets achieve near-peak performance at smaller patch sizes (e.g., 6 or 8), possibly due to their higher spectral redundancy or smaller spatial variances. While for the UH dataset, accuracy peaks early and fluctuates, indicating that this dataset may have a lower dependence on large spatial contexts. The SA dataset exhibits the highest overall accuracy (OA) (99.7%), demonstrating the model's ability to effectively utilize both spectral and spatial information in highly correlated data. The HC dataset has the lowest starting accuracy (90.1%), indicating a higher sensitivity to inadequate spatial context at smaller patch sizes as shown in Fig. 7. The results reinforce the importance of patch size in HSIC. The need to adaptively select patch sizes based on dataset characteristics. The ability of the model to maintain high performance across datasets with diverse spatial and spectral characteristics.

F. Feature Representation

t-distributed stochastic neighbor embedding (t-SNE) is a dimensionality reduction technique employed to visualize high-dimensional data by projecting it into a lower-dimensional space, as shown in Fig. 8. The method preserves the pairwise similarities between data points, ensuring that points in the high-dimensional space that are similar to one another remain close in the lower-dimensional projection. This is achieved through an objective function based on KL divergence, which quantifies the dissimilarity between probability distributions in the high- and low-dimensional spaces. The algorithm iteratively adjusts the positions of data points in the lower-dimensional space to minimize this divergence, thereby retaining meaningful spatial relationships.

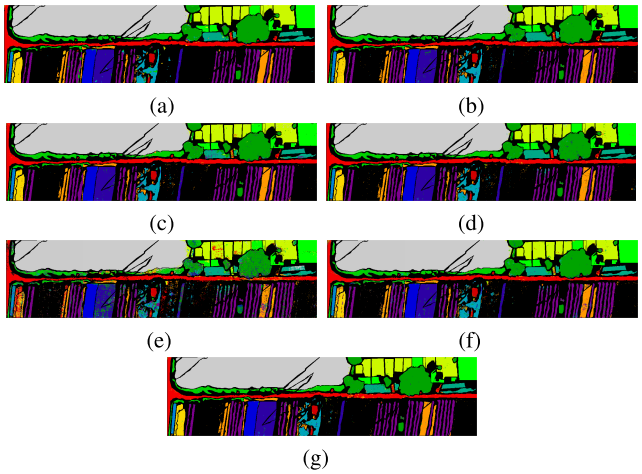


Fig. 9. Predicted ground-truth maps for the HC dataset, showcasing the performance of different state-of-the-art methods. The proposed BLM-KD model demonstrates superior accuracy and clearer delineation of land-cover classes compared to (a) AGCN, (b) SST, (c) HViT, (d) WFormer, (e) Mamba, (f) WMamba, and (g) BLM-KD.

In the context of this study, t-SNE is utilized to evaluate the feature representations learned by the BLM-KD model. Fig. 8 presents t-SNE visualizations for the HH, HC, PU, SA, and UH datasets. These visualizations illustrate the student model's capacity to effectively differentiate between distinct classes, thereby validating the efficacy of its feature learning capabilities across various datasets. The clear clustering observed in the t-SNE plots suggests that the model accurately captures the underlying structure of the data, contributing to improved classification performance.

VI. Qualitative and Quantitative Results

This section presents a comprehensive analysis of the qualitative and quantitative results achieved by the BLM-KD in comparison to several state-of-the-art methods. The comparative methods, including the graph attention CNN (AGCN), Vision Transformer, hybrid Vision Transformer (HViT), Waveformer (WFormer), Spectral Mamba, and WaveMamba, were implemented using Python with TensorFlow/Keras libraries. All experiments were conducted in a Linux environment on a high-performance portable machine equipped with an NVIDIA RTX 4060 GPU, 64 GB of RAM, and an Intel i9 processor. For fairness, a consistent experimental setup was adopted across all methods, employing an 8×8 patch size, a training sample proportion of 25%, a learning rate of 0.001, and a batch size of 56.

The results presented in Table II and the corresponding predicted ground-truth maps in Fig. 9 provide a detailed comparison of the proposed BLM-KD method against state-of-the-art models, including AGCN [73], SST [74], HViT [75], WFormer [26], Spectral Mamba (Mamba) [76], and WaveMamba (WMamba) [35]. The evaluation metrics considered include class-wise accuracy, κ, OA, and average accuracy (AA).

The proposed BLM-KD model outperforms all other methods in most classes, demonstrating its robustness across diverse

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TABLE II

PERFORMANCE COMPARISON OF VARIOUS STATE-OF-THE-ART METHODS FOR HSIC ON THE HC DATASET

Class	AGCN	SST	HViT	WFormer	Mamba	WMamba	BLM-KD
Strawberry	99.3517	99.0253	98.7616	98.4531	94.0182	98.0194	98.9359
Cowpea	95.9391	98.3651	98.9012	97.3279	87.9405	93.4604	98.2332
Soybean	97.8806	95.0418	97.1611	98.5222	66.1481	94.9445	98.0167
Sorghum	97.6466	99.2528	99.6264	99.6264	97.2730	98.9166	99.5890
Water spinach	95.5000	93.8333	94.6666	94.1666	52.3333	86.6666	95.3333
Watermelon	80.8914	94.1306	90.4236	90.2912	45.4545	75.1544	95.5251
Greens	91.3617	95.4268	79.7764	97.3238	80.0813	92.7506	96.8834
Trees	94.1929	96.4512	96.5735	95.4277	70.0634	94.0816	94.8270
Grass	99.3029	96.8525	95.8386	97.7186	65.5893	93.7896	94.8880
Red roof	98.8779	99.4674	98.6877	99.4294	95.4925	98.9729	99.1441
Gray roof	99.3968	98.8764	97.2324	97.3388	91.5316	97.5044	98.3560
Plastic	97.2282	87.7173	95.2173	94.0217	29.4565	79.7826	95.9239
Bare soil	84.9714	91.3558	92.8916	91.6849	62.3299	78.8503	89.6007
Road	97.3599	95.6250	97.5969	97.2844	82.8879	94.7737	97.6508
Bright object	91.5492	89.2605	91.0211	92.0774	57.3943	78.5211	89.2605
Water	99.5968	99.6366	99.7639	99.7665	99.0185	99.6472	99.8912
κ	96.9637	96.4557	96.2835	96.5300	84.4072	95.0004	**97.4907**
OA	96.4061	96.8262	96.6802	96.8891	86.7029	95.7286	**97.8565**
AA	94.0654	94.6449	94.2587	94.2788	73.5633	90.9897	**95.7537**

TABLE III

PERFORMANCE COMPARISON OF VARIOUS STATE-OF-THE-ART METHODS FOR HSIC ON THE HH DATASET

Class	AGCN	SST	HViT	WFormer	Mamba	WMamba	BLM-KD
Red roof	96.6666	99.0170	99.4586	99.4729	94.8005	99.1025	99.5726
Road	96.2414	97.6651	95.3872	93.0523	69.5330	90.3189	94.4760
Bare soil	94.8583	98.1578	98.7627	98.2036	90.2392	97.3787	98.8910
Cotton	99.7709	99.7881	99.5774	99.9338	99.6104	99.5982	99.8946
Cotton firewood	94.1138	91.8944	98.2952	80.7011	42.9398	92.1196	92.9559
Rape	95.0267	97.9666	98.9272	99.0484	95.8795	98.9272	99.4748
Chinese cabbage	97.4858	96.4404	96.9216	97.2618	80.8911	95.8347	96.0089
Pakchoi	76.8623	88.4065	87.9131	91.5145	46.5219	77.3063	92.3038
Cabbage	97.6150	99.3344	99.5378	99.2050	97.2823	98.4285	98.9092
Tuber mustard	92.5770	93.1418	94.1584	91.1892	72.6964	94.1746	98.7090
Brassica parachinensis	85.0917	97.7120	96.4227	95.3876	68.9122	93.5355	94.1710
Brassica chinensis	83.5157	95.6444	94.2148	93.5447	31.9633	85.7717	97.3866
Small Brassica chinensis	98.4361	96.3479	97.4497	95.8592	63.6751	92.1539	95.8681
Lactuca sativa	97.1179	96.0576	95.5138	95.7857	82.5992	97.6073	98.1783
Celtuce	83.8323	95.0099	98.0039	98.2035	80.4391	78.4431	94.2115
Film covered lettuce	96.7226	96.1167	98.9534	98.5954	82.4290	93.6105	98.3200
Romaine lettuce	95.7475	97.2757	97.8737	96.1461	90.2990	95.3488	96.9435
Carrot	97.8855	91.6044	96.9527	97.8855	86.1940	95.9577	95.3358
White radish	92.1946	96.6942	97.2451	95.1331	82.0247	95.0872	97.7961
Garlic sprout	95.1233	91.6810	96.0413	94.6643	74.3545	86.0585	94.6069
Broad bean	90.8132	76.9578	92.0180	87.5000	12.6506	61.2951	93.0722
Tree	85.2970	97.5247	97.5742	99.1089	46.7821	88.2178	97.3267
κ	95.6726	97.3688	97.8958	97.3589	84.4208	95.9923	**98.0922**
OA	96.5750	97.9182	98.3335	97.9135	87.7546	96.8295	**98.4913**
AA	92.8634	95.0199	95.6910	95.3362	72.3962	91.1944	**96.5642**

land-cover categories. For instance, in the Sorghum class, BLM-KD achieves an accuracy of 99.5890%, closely matching the performance of WFormer (99.6264%) but exceeding all others. Similarly, in the Watermelon class, BLM-KD significantly surpasses AGCN, HViT, and WFormer with an accuracy of 95.5251%, highlighting its superior feature extraction capability. Furthermore, for challenging categories such as plastic and bare soil, BLM-KD achieves notable improvements with accuracies of 95.9239% and 89.6007%, respectively, outperforming Mamba and WMamba by substantial margins. BLM-KD demonstrates a consistent advantage in aggregated metrics. BLM-KD achieves the highest κ (97.4907%), OA (97.8565%), and AA (95.7537%), reflecting its effectiveness in capturing spatial–spectral features and generalizing across classes. Compared to Mamba, which suffers from lower accuracies in most classes due to its limited spectral–spatial integration, BLM-KD leverages enhanced KD and a balanced learning mechanism, leading to superior performance.

The predicted ground-truth maps in Fig. 9 provide qualitative evidence of the improvements. BLM-KD produces cleaner and more accurate classifications, particularly in regions with complex spectral variability, as seen in the water and strawberry classes. In contrast, AGCN and SST show visible misclassifications in these areas, while WMamba exhibits artifacts, particularly for low-contrast classes like bright objects and watermelon. The visual clarity in BLM-KD's outputs highlights its ability to prioritize and fuse spatial and spectral information effectively.

The performance comparison on the HH dataset is summarized in Table III. The table reports the classification accuracy for each class in the dataset. The results indicate that the proposed BLM-KD model outperforms all other methods, including AGCN, SST, HViT, WFormer, Mamba, and WMamba, across most land-cover classes. Notably, BLM-KD achieved the highest κ (98.0922), OA (98.4913), and AA (96.5642), demonstrating its superiority in both overall performance and class-wise accuracy. For instance, in the "cotton" class, BLM-KD reached a classification accuracy of 99.8946%, outperforming the next best method, WFormer, which achieved 99.9338%. Furthermore, BLM-KD delivered the most significant improvement in the "broad bean" class,

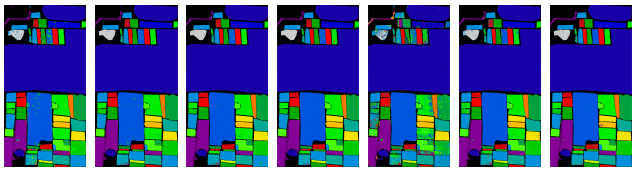


Fig. 10. Predicted ground-truth maps for the HH dataset, showcasing the performance of different state-of-the-art methods. The proposed BLM-KD model demonstrates superior accuracy and clearer delineation of land-cover classes.

where other models struggled, such as Mamba, which achieved only 12.6506%. This highlights the robustness of the BLM-KD model, particularly in challenging cases with complex or poorly defined land-cover classes.

Fig. 10 visually complements the quantitative results in Table III by showcasing the predicted ground-truth maps for the HH dataset. The results indicate that BLM-KD provides clearer and more accurate delineation of land-cover classes when compared to the other methods. The predicted maps illustrate the ability of BLM-KD to better capture fine-grained details and boundaries between different land-cover types, reflecting the higher performance metrics achieved in Table III. The proposed model's superior accuracy is particularly evident in areas with complex land-cover patterns, where other models fail to adequately distinguish between adjacent classes. These visual results reinforce the strong performance of BLM-KD, validating its potential for more precise and reliable HSIC.

Table IV presents a performance comparison on the PU dataset, evaluated across several metrics. Among the methods evaluated, the proposed BLM-KD model consistently outperforms the others in all key metrics, achieving the highest values for κ (99.4608), OA (99.5932), and AA (99.2309). The superior performance of BLM-KD can be attributed to its enhanced capability in delineating land-cover classes, as reflected in its improved classification accuracy across diverse classes such as asphalt, meadows, gravel, and shadows. Notably, BLM-KD achieves perfect accuracy (100%) in several classes, including painted metal sheets and bare soil, further emphasizing its robustness and efficacy in handling challenging datasets. In

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TABLE IV

PERFORMANCE COMPARISON OF VARIOUS STATE-OF-THE-ART METHODS FOR HSIC ON THE PU DATASET

Class	AGCN	SST	HViT	WFormer	Mamba	WMamba	BLM-KD
Asphalt	99.2460	98.4620	98.7032	99.1556	95.9891	99.2460	99.7587
Meadows	99.9249	99.7640	99.7640	99.8605	98.1767	99.8712	100
Gravel	95.1401	90.8484	92.3736	91.7063	77.4070	93.6129	96.8541
Trees	99.2819	99.1514	99.0861	99.4778	96.4751	98.7597	99.2819
Painted metal sheets	100	99.5542	100	99.8514	100	100	100
Bare Soil	100	99.8806	99.7215	99.8011	95.7836	99.8011	100
Bitumen	98.6466	98.6466	98.4962	98.3458	89.0225	97.5939	99.5488
Self-Blocking Bricks	99.8913	95.4372	95.9261	94.5138	87.7240	95.8718	98.4790
Shadows	99.5780	98.7341	98.5232	98.9451	96.4135	98.7341	99.1561
κ	98.4672	98.2221	98.4017	98.3953	93.70481	98.5997	**99.4608**
OA	98.5979	98.6581	98.7937	98.7890	95.2496	98.9433	**99.5932**
AA	98.3010	97.8309	98.0660	97.9619	92.9990	98.1656	**99.2309**

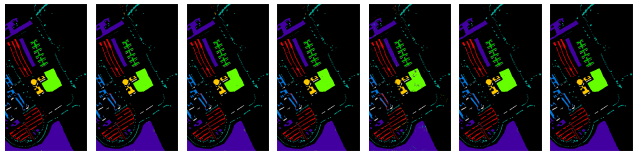


Fig. 11. Predicted ground-truth maps for the PU dataset, showcasing the performance of different state-of-the-art methods. The proposed BLM-KD model demonstrates superior accuracy and clearer delineation of land-cover classes.

comparison, models such as AGCN, SST, HViT, and WFormer show competitive performance, with OA values ranging from 98.5% to 98.8%. However, they lag behind BLM-KD in terms of both κ and AA, indicating that while they provide relatively high OA, they struggle more in terms of class-wise performance, particularly in the more challenging categories like self-blocking bricks and gravel. Mamba and WMamba, although performing reasonably well, exhibit relatively lower performance in several classes, especially in terms of OA and AA. Specifically, Mamba achieves an OA of 95.2496% and an AA of 92.9990%, significantly trailing behind BLM-KD in these metrics.

Fig. 11 provides a visual representation of the predicted ground-truth maps for the PU dataset, highlighting the performance of the various methods. The BLM-KD model, shown in the last subfigure, not only achieves the highest accuracy but also demonstrates superior ability to delineate distinct land-cover classes with clearer boundaries and fewer misclassifications. This is in stark contrast to other models, where the predicted boundaries are often blurred or exhibit more errors. The visual comparison reinforces the quantitative results presented in the table, underscoring the effectiveness of the proposed method in achieving both high accuracy and precise class delineation.

The performance comparison on the SA dataset, as presented in Table V, shows a clear distinction in accuracy across different land-cover classes. Among the models evaluated, the BLM-KD model outperforms all other models. For example, the BLM-KD model achieves an OA of 99.8115%, AA of 99.8691%, and κ of 99.79017%, surpassing other models in all these metrics. The accuracy of the individual classes, such as "broccoli green weeds 2" and "fallow rough plow," demonstrates the BLM-KD model's strong performance, as it achieves 100% classification accuracy in several instances, compared to other models like Mamba and WMamba, which show slightly lower accuracies for certain classes. The model's

TABLE V

PERFORMANCE COMPARISON OF VARIOUS STATE-OF-THE-ART METHODS FOR HSIC ON THE SA DATASET

Class	AGCN	SST	HViT	WFormer	Mamba	WMamba	BLM-KD
Broccoli green weeds 1	99.9003	100	100	99.8007	99.9003	100	100
Broccoli green weeds 2	100	99.8926	100	100	99.8926	100	99.9463
Fallow	100	100	99.8987	100	98.5829	99.6963	100
Fallow rough plow	100	100	100	99.5695	96.7001	99.8565	100
Fallow smooth	99.8506	100	99.5519	99.4772	98.8797	99.1038	99.4025
Stubble	100	100	100	100	99.8989	100	100
Celery	100	99.8324	99.8324	100	99.3854	99.9441	100
Grapes untrained	99.7870	98.1547	98.8289	98.3676	94.7657	99.8403	99.6983
Soil vinyard development	100	100	100	100	99.8710	100	100
Corn senesced green weeds	99.6949	99.8779	100	99.7559	99.3288	98.9017	99.7559
Lettuce romaine 4wk	99.8127	99.8127	100	100	98.8764	100	100
Lettuce romaine 5wk	100	100	100	100	99.7923	100	100
Lettuce romaine 6wk	100	100	99.7816	100	100	99.5633	100
Lettuce romaine 7wk	99.2523	99.8130	100	99.6261	99.2523	99.4392	99.6261
Vinyard untrained	91.3043	96.8904	97.3582	96.9455	79.5542	98.7066	99.4771
Vinyard vertical trellis	100	100	100	100	98.8925	100	100
κ	98.5958	99.0701	99.2881	99.0907	95.2911	99.6049	**99.79017**
OA	98.7400	99.1649	99.3607	99.1834	95.7768	99.6452	**99.8115**
AA	99.3501	99.6421	99.7032	99.5964	97.7233	99.6907	**99.8691**

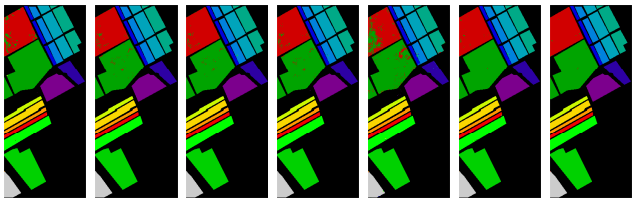


Fig. 12. Predicted ground-truth maps for the SA dataset, showcasing the performance of different state-of-the-art methods. The proposed BLM-KD model demonstrates superior accuracy and clearer delineation of land-cover classes.

capacity to handle complex and varied land-cover classes, particularly in challenging areas such as "vineyard untrained," where BLM-KD achieves a notable 99.4771%, is also evident.

Fig. 12 visually complements the quantitative results by showcasing predicted ground-truth maps for each model on the SA dataset. Each subfigure represents the output of a different model, with the BLM-KD model exhibiting the most refined delineation of land-cover boundaries and the most accurate classification of diverse classes. Compared to other models, BLM-KD's output shows clearer and more precise segmentation, especially in difficult-to-classify regions, such as "vineyard untrained" and "broccoli green weeds 2." This enhanced clarity highlights the superior performance of BLM-KD in accurately identifying land-cover types and distinguishing between them, which is critical for applications in remote sensing and precision agriculture.

As shown in Table VI, notably, the proposed BLM-KD model achieves the highest OA (99.4011%), AA (99.3066%), and κ (99.3526%), demonstrating its superior capability in classifying HSIs. Class-specific accuracies reveal consistent outperformance by BLM-KD across several critical classes, such as synthetic grass, soil, and commercial, where it achieves perfect or near-perfect scores. The SST and HViT models also exhibit strong performance, with OAs of 99.1483% and 99.2548%, respectively. However, these methods fall short in consistently classifying challenging classes, such as parking lot 2 and residential, compared to BLM-KD. Legacy methods, such as AGCN and Mamba, while competitive, exhibit limitations in robustness, particularly for heterogeneous classes like parking lot 2, where their accuracies drop to 89.7435% and 72.2222%, respectively.

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TABLE VI

PERFORMANCE COMPARISON OF VARIOUS STATE-OF-THE-ART METHODS FOR HSIC ON THE UH DATASET

Class	AGCN	SST	HViT	WFormer	Mamba	WMamba	BLM-KD
Healthy grass	97.7635	98.5623	99.2012	99.3610	97.7635	97.7635	98.5623
Stressed grass	98.7240	99.8405	99.5215	99.8405	99.3620	99.5215	99.8405
Synthetic grass	100	99.7126	100	100	98.2758	99.7126	100
Trees	100	99.8392	99.5176	100	99.8392	99.3569	99.6784
Soil	100	100	100	100	99.3558	99.5169	100
Water	91.9753	99.3827	100	100	92.5925	99.3827	100
Residential	99.2113	97.1608	97.9495	98.1072	93.8485	96.0567	98.1072
Commercial	96.3022	99.6784	99.8392	99.5176	89.3890	99.3569	100
Road	94.0894	100	99.2012	100	95.3674	99.0415	100
Highway	99.0228	99.8371	99.8371	99.8371	89.7394	98.8599	99.6742
Railway	98.0582	99.1909	99.1909	100	98.0582	99.6763	99.8381
Parking Lot 1	93.3549	98.8654	99.3517	98.8654	89.9513	98.3792	99.0275
Parking Lot 2	89.7435	92.7350	93.5897	93.1623	72.2222	94.4444	94.8717
Tennis Court	100	99.5327	99.5327	100	99.0654	100	100
Running Track	100	100	100	99.6969	99.6969	99.3939	100
κ	97.3666	99.0792	99.1943	99.3525	94.5311	98.6333	99.3526
OA	97.5648	99.1483	99.2548	99.4011	94.9434	98.7358	99.4011
AA	97.2163	98.9558	99.1155	99.2258	94.3018	98.6975	99.3066

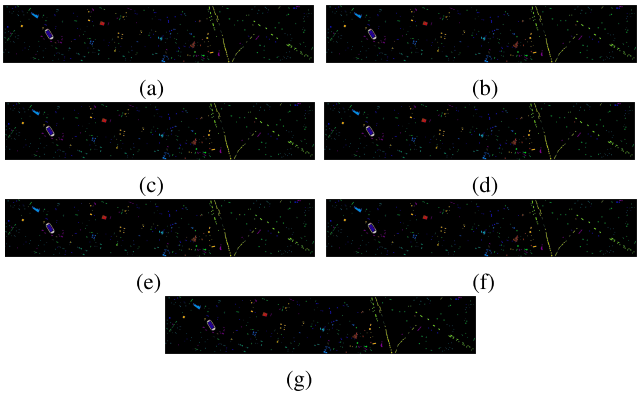


Fig. 13. Predicted ground-truth maps for the UH dataset, showcasing the performance of different state-of-the-art methods. The proposed BLM-KD model demonstrates superior accuracy and clearer delineation of land-cover classes. (a) AGCN. (b) SST. (c) HViT. (d) WFormer. (e) Mamba. (f) WMamba. (g) BLM-KD.

Fig. 13 presents predicted ground-truth maps for the UH dataset, visually comparing the classification outputs of the aforementioned methods. The BLM-KD model [Fig. 13(g)] showcases superior spatial delineation and reduced misclassification, particularly in complex regions with overlapping or transitional land-cover classes. For instance, areas corresponding to residential and commercial classes are more distinctly segmented in the BLM-KD output, indicating the model's capability to handle spectral variability and spatial heterogeneity. The WFormer model [Fig. 13(d)] also produces high-quality results but exhibits slight over-smoothing in regions with intricate spatial patterns. Similarly, SST [Fig. 13(b)] demonstrates sharp transitions but suffers from minor artifacts in class boundaries. By contrast, AGCN [Fig. 13(a)] and Mamba [Fig. 13(e)] outputs highlight the limitations of earlier architectures, with visible misclassifications and boundary inconsistencies, particularly in densely populated classes like running track and parking lot 1.

VII. COMPUTATIONAL COST OF COMPARATIVE METHODS

To evaluate the computational efficiency of the proposed BLM-KD framework against existing methods, we compare training time, inference time, and the number of parameters for each model. These factors are critical for practical deployment,

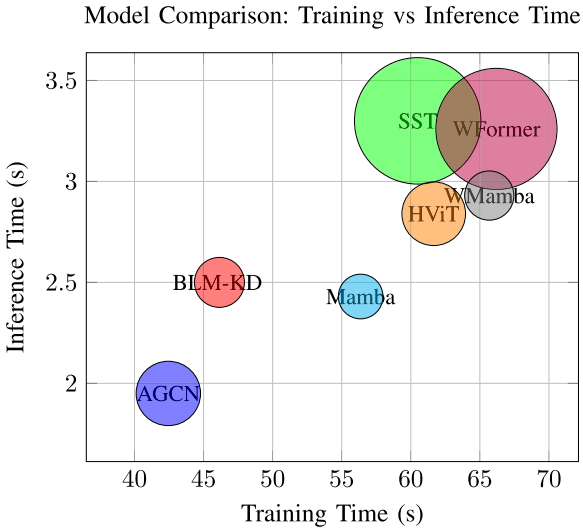


Fig. 14. Computational cost of comparative models: training versus inference time (bubble size = parameters in M). Bracketed values indicate relative FLOPs per single-sample inference (rFLOPs; BLM-KD= 1.00): BLM-KD [1.00], AGCN [0.78], SST [1.32], HViT [1.14], WFormer [1.30], Mamba [0.97], and WMamba [1.17].

particularly in resource-constrained environments or time-sensitive applications. Fig. 14 presents a bubble plot where the x-axis denotes training time, the y-axis represents inference time, and the size of each bubble corresponds to the number of parameters.

As illustrated in Fig. 14, BLM-KD achieves a strong balance between computational efficiency and model complexity. Despite having a moderately sized parameter count (0.66 M), it demonstrates competitive performance with one of the lowest training times and a favorable inference time among all compared methods. AGCN and Mamba show slightly lower inference times, but this comes at the cost of reduced model capacity or longer training durations. SST and WFormer, while offering larger representational capacity, incur significantly higher computational costs in both training and inference. Similarly, WMamba and HViT also demand greater resources without a proportional gain in efficiency. These results underscore the advantage of BLM-KD in delivering an efficient solution that minimizes computational overhead while retaining sufficient model complexity.

VIII. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This work presented the BLM-KD framework for HSIC, addressing key limitations in existing methods related to tokenization, computational inefficiency, and spectral–spatial modeling. Unlike prior approaches that rely on token-based segmentation, BLM-KD introduces a *tokenization-free BLM* mechanism that learns directly from raw hyperspectral data. This preserves spectral continuity and spatial coherence while reducing preprocessing complexity and avoiding fragmentation artifacts. To effectively model spatial–spectral dependencies, BLM-KD integrates a structured SSM that captures both local and long-range contextual relationships through dynamic state transitions, all while maintaining linear computational complexity. Additionally, an *adaptive KD* strategy enables efficient training of a lightweight student model

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by selectively transferring informative features from a more expressive teacher model. A temperature-controlled distillation schedule ensures robust generalization while minimizing computational overhead. Extensive experiments conducted on multiple benchmark hyperspectral datasets validate the effectiveness of the proposed framework. BLM-KD consistently outperforms state-of-the-art baselines in classification accuracy while achieving superior scalability and efficiency.

While BLM-KD advances HSIC significantly, several promising avenues for further exploration are identified. Although the KD framework balances accuracy and efficiency, further research into advanced model compression techniques, such as pruning and quantization, could optimize the student model for real-time deployment in resource-constrained environments. The framework's resilience to varying noise levels and atmospheric conditions in hyperspectral data warrants further study. Incorporating domain adaptation techniques could enhance the robustness of the model under diverse environmental and imaging conditions. BLM-KD could be extended to fuse data from multiple modalities, such as LiDAR and thermal imaging, alongside hyperspectral data. Such integration could unlock deeper insights and improve classification performance in complex environments.

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